

Digitally Enabled Quality Improvement Intervention and LDL-C Control in Atherosclerotic Cardiovascular Disease

The SAPPHIRE-LDL Cluster Randomized Clinical Trial

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IMPORTANCE Bridging the low-density lipoprotein cholesterol (LDL-C) care gap in patients with established atherosclerotic cardiovascular disease (ASCVD) is challenging. Few quality improvement (QI) interventions have successfully improved patient care.

OBJECTIVE To evaluate the impact of a digitally enabled, multifaceted, QI intervention on the control of LDL-C concentration in patients with ASCVD.

DESIGN, SETTING AND PARTICIPANTS This was a pragmatic, 2-arm, cluster randomized clinical trial. A total of 28 clusters (public or private outpatient clinics) in Brazil were randomized to receive a digitally enabled QI intervention or routine practice (control). Adult patients (≥ 18 years) with established ASCVD were enrolled between November 2023 and December 2024 and followed up for 6 months.

INTERVENTION The intervention comprised a structured, digitally enabled QI strategy integrating previsit screening, electronic clinical decision support algorithms, audit and feedback mechanisms, and targeted clinician and patient engagement tools embedded within routine workflows to support lipid monitoring, treatment intensification, and adherence.

MAIN OUTCOMES AND MEASURES The primary outcome was mean LDL-C level at 6 months. Secondary outcomes included relative LDL-C change, achievement of LDL-C targets (< 50 mg/dL and $\geq 50\%$ reduction; to convert to millimoles per liter, multiply by 0.0259), and prescription of lipid-lowering therapies. All analyses were performed following the intention-to-treat principle and accounted for the cluster design by using a generalized estimating equations model.

RESULTS Among 1465 enrolled patients (mean [SD] age, 62.3 [10.1] years; 894 male [61.0%]; 714 intervention; 751 control), mean (SD) LDL-C concentration at 6 months was 76.3 (37.5) mg/dL in the intervention group and 85.6 (37.1) mg/dL in the control group. The adjusted mean difference was -6.62 mg/dL (95% CI, -11.11 to -2.13 mg/dL; $P = .004$). Patients in the intervention group were more likely to achieve an LDL-C concentration less than 50 mg/dL (23.5% vs 13.4%; odds ratio, 1.86; 95% CI, 1.30-2.65) and a 50% or greater reduction in LDL-C concentration (18.6% vs 13.4%; odds ratio, 1.76; 95% CI, 1.27-2.43). Prescription of intensive and combination lipid-lowering therapy was significantly higher in the intervention group. No significant differences in major cardiovascular events were observed.

CONCLUSIONS AND RELEVANCE In this pragmatic cluster randomized clinical trial, a digitally enabled, multifaceted strategy produced modest LDL-C reduction and increased treatment intensification among patients with ASCVD. However, a proportion of patients had LDL-C levels that remained above recommended LDL-C targets, underscoring that the strategy tested was only partially effective in eliminating the residual care gap.

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Cardiovascular disease remains the leading cause of morbidity and mortality globally and in Brazil.¹ Low-density lipoprotein cholesterol (LDL-C) plays a direct causal role in atherosclerotic cardiovascular disease (ASCVD).²⁻⁴ Accordingly, contemporary guidelines recommend intensive lipid-lowering therapy for secondary prevention, targeting at least a 50% reduction in LDL-C concentration and absolute levels below 50 to 55 mg/dL (to convert to millimoles per liter, multiply by 0.0259) in patients with established ASCVD.⁵⁻⁷

Despite these recommendations, implementation in routine practice remains suboptimal. Data from large international registries, including the EU-Wide Cross-Sectional Observational Study of Lipid-Modifying Therapy Use in Secondary and Primary Care (DA VINCI)⁸ and Study of Approaches to the Management of Patients With Hypercholesterolemia in Diverse European Practice (SANTORINI),⁹ consistently demonstrate low rates of LDL-C goal attainment and limited use of combination therapy despite widespread statin prescription, particularly relevant in settings with constrained resources.^{1,10}

Quality improvement (QI) interventions have shown inconsistent effects on LDL-C control; for example, the Brazilian Intervention to Increase Evidence Usage (BRIDGE Cardiovascular Prevention) intervention increased statins prescription by 9% without measurable LDL-C improvements.¹¹

Digitally enabled, multifaceted QI strategies that embed guideline pathways into clinical workflows may facilitate treatment intensification and improve lipid control, yet randomized evidence supporting this approach in middle-income settings is limited. To address this gap, we conducted the Systematic Assessment of a Pragmatic, Practice-Based, Health Care Intervention for Reducing Elevated LDL (SAPPHIRE-LDL) pragmatic cluster randomized clinical trial to evaluate whether a digitally enabled QI intervention improves LDL-C levels, LDL-C target attainment, and use of intensive and combination lipid-lowering therapy among patients with established ASCVD in Brazil.

Methods

Study Design

The study was approved by the Einstein Hospital Israelita Human Research Ethics Committee, with local approvals obtained. The trial protocol and prerandomization results were published elsewhere¹² and appear in [Supplement 1](#). The statistical analysis plan appears in [Supplement 2](#). Consent was obtained at the cluster level for participation, randomization, intervention assignment, and aggregated data collection in the study and at the individual level, for receiving the intervention and for data collection (eAppendix 1 in [Supplement 3](#)). This report followed the Consolidated Standards of Reporting Trials (CONSORT) 2010 reporting guidelines and the Consolidated Standards of Reporting Trials Extension (CONSORT Extension) for cluster randomized clinical trials.

SAPPHIRE-LDL was a pragmatic, 2-arm, open-label, cluster randomized clinical trial with intention-to-treat analysis designed to assess the effect of a digitally enabled QI intervention on the control of LDL-C control, lipid-lowering therapy

Key Points

Question Does a digitally enabled, multifaceted, quality improvement intervention improve low-density lipoprotein cholesterol (LDL-C) control among patients with established atherosclerotic cardiovascular disease?

Findings In this cluster randomized clinical trial including 1465 patients across 28 outpatient clinics, the intervention resulted in a statistically significant reduction in mean LDL-C concentration at 6 months and higher rates of LDL-C target attainment and intensive combination lipid-lowering therapy compared with usual care.

Meaning Results of this study show that digitally enabled, multifaceted, quality improvement strategies can improve lipid management and reduce evidence-practice gaps in secondary cardiovascular prevention in real-world settings.

prescription, and cardiovascular events in patients with ASCVD.

Clusters and Patients

Clusters were outpatient clinics from public and private hospitals from all regions in Brazil. Both teaching and nonteaching units were eligible. We identified clusters through previous registries, our research network, and medical societies. We enrolled consecutive male and female patients older than 18 years who had ASCVD from November 2023 to December 2024. Detailed eligibility criteria are shown in [Supplement 1](#) and eAppendix 2 in [Supplement 3](#). We obtained information about race and ethnicity as self-reported by the patients and used categories that aligned with the Brazilian Institute of Geography and Statistics classification. Races included Black, East Asian, Indigenous, multiracial, and White.

Randomization and Allocation Concealment

Clusters were randomly allocated (1:1) to a multifaceted digitally enabled QI strategy (intervention group) or to routine practice (control group) in 3 installments by a statistician using a central web-based randomization system. Randomization was stratified according to the observed primary outcome during the prerandomization phase (eAppendix 3 in [Supplement 3](#)).

Blinding

Patients and investigators were not blinded to the allocation of treatment. Statisticians were blinded to the nature of the intervention.

Digitally Enabled QI Intervention

The QI intervention published elsewhere,¹² operated through a single integrated platform (epScience/epYou [epHealth]) across 4 touchpoints of the outpatient encounter (eFigure 1 and eAppendix 4 in [Supplement 3](#)). Before each visit, a coordinator, nurse, or physician completed a standardized checklist of cardiovascular comorbidities, current lipid-lowering treatment (LLT), and lipid values, and obtained a point-of-care LDL-C testing (POCT) (Afinion 2 [Abbott]); venous sampling was used when POCT were inconclusive. During the consultation, these data populated an electronic clinical decision support algorithm aligned with contemporary Brazilian Society of

Cardiology guidelines⁶; if the clinician's intended decision diverged from the algorithm, a real-time prompt invited reconsideration while preserving prescribing autonomy. Between visits, patients used a mobile application (epYou [epHealth]) delivering educational content, medication reminders, longitudinal lipid visualization, and bidirectional communication with the care team. At the cluster level, periodic audit and feedback sessions using structured reports that summarized lipid control and prescribing patterns to identify barriers and set goals, complemented by tailored clinician educational materials.

Beyond initial site training and a brief previsit checklist embedded in the routine consultation, all clinician- and patient-facing tools were delivered automatically through the digital platform, with no dedicated study staff at the point of care. Usual-care clusters continued routine practice with general electronic educational materials only.

Data Collection

Data were collected using a secure, web-based electronic data capture platform (epScience/epYou [epHealth]), operating across both study arms at baseline, 3 months, 6 months, and during on-demand visits. Study coordinators entered data at each site. Point-of-care lipid results were automatically integrated from Afinion 2 (Abbott) devices; laboratory results were entered manually when applicable.

End Points

The primary end point was the LDL-C concentration measured at the end of the follow-up (sixth month). Secondary outcomes included LDL-C concentration and prescribed LLT outcomes, as well as barriers for drug prescription at the system (cluster) level and physician level adherence to prescribed LLT, barriers for drug adherence, and statin intolerance.

Exploratory end points included major cardiovascular events and mortality. eAppendix 5 in Supplement 3 contains detailed end point definitions.

Sample Size

Based on the prerandomization observational phase (observed intracluster correlation coefficient [ICC] of 0.028; SD 40 mg/dL), 28 clusters provide us 80% power to detect a 15 mg/dL absolute LDL-C difference at 6 months, assuming a conservative 0.10 ICC (eAppendix 6 in Supplement 3).

Statistical Analysis

Continuous variables are presented as mean (SD) or median (IQR) and categorical variables as counts and percentages. Analyses followed the intention-to-treat principle.

The primary outcome (LDL-C level at 6 months) was estimated using generalized estimating equations with an exchangeable correlation structure, adjusted for baseline LDL-C and prerandomization stratification factors (site-level median LDL-C concentration) and reported as adjusted mean differences with 95% CIs.

Secondary lipid and prescription outcomes were analyzed similarly with binary outcomes reported as population-average odds ratios (ORs) with 95% CI.

Time-to-event outcomes were analyzed with frailty Cox proportional hazards models with a cluster-level random effect (hazard ratios [HRs] with 95% CIs). Prespecified sensitivity analyses assessed robustness to covariate adjustment, alternative LDL-C ascertainment, missing data handling, and LDL-C calculation approaches; subgroup analyses used interaction terms. A 2-sided *P* value <.05 was considered significant. Analyses used the latest version of R (R Project for Statistical Computing).

Results

Of 56 clusters invited, 28 met criteria with an average 50 patients each for randomization (Figure 1). A total of 1465 patients were enrolled (mean [SD] age, 62.3 [10.1] years; 571 female [39.0%]; 894 male [61.0%]; 714 intervention; 751 control). Participants self-identified the following races and ethnicities: 172 Black (11.7%), 8 East Asian (0.5%), 6 Indigenous (0.4%), 554 multiracial (37.8%), and 725 White (49.5%). Six-month follow-up was completed for 634 of 714 patients (88.8%) and 713 of 751 patients (94.9%) in the intervention and control groups, respectively, and complete LDL-C data were available for 629 of 714 patients (88.1%) and 700 of 751 patients (93.2%) in the intervention and control groups, respectively, constituting the prespecified complete-case analysis population. Reasons for missing data are detailed in eTable 1 in Supplement 3.

Baseline Characteristics

Baseline cluster and patient characteristics are presented in Table 1. From the included clusters, 24 (85.7%) were cardiology-predominant sites, 5 (17.8%) were family health strategies, 17 (60.7%) were teaching hospitals, and the median (IQR) volume of patients seen was approximately 80 (50-200) patients per month. A total of 1214 patients (82.9%) had coronary artery disease (998 [68.1%] with prior myocardial infarction), 251 (17.1%) had prior ischemic stroke or transient ischemic attack, 160 (10.9%) had peripheral vascular disease, 1240 (84.6%) had a history of hypertension, and 592 (40.4%) had diabetes. The mean (range) number of patients in each cluster was 52 (10-120), and the median was 42 patients.

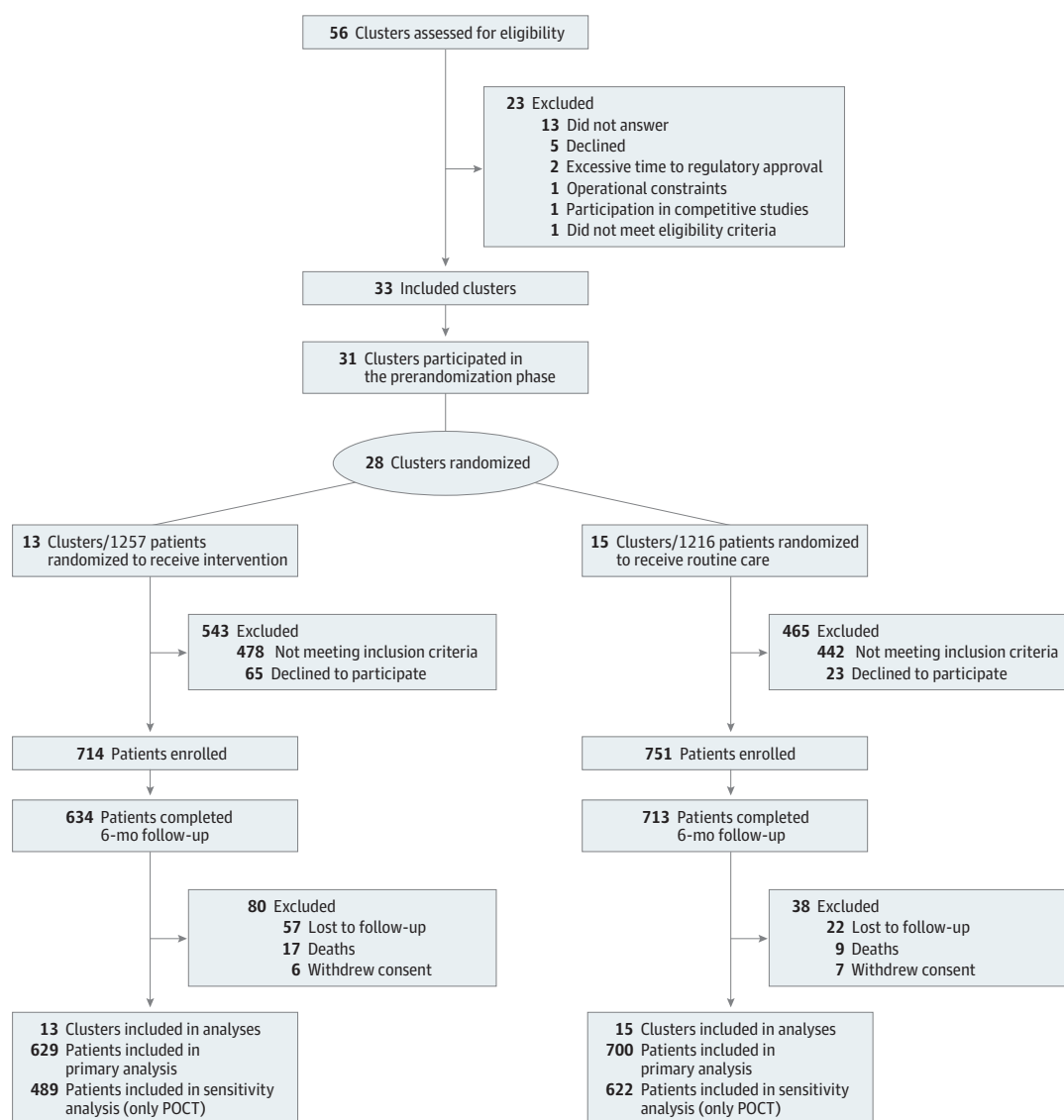
Adherence to the SAPHIRE Intervention

Adherence to QI components was high. The decision support tool was used in 1683 of 1736 opportunities (96.9%); 452 of 714 patients (63.0%) in the intervention group and 402 of 751 patients (54.0%) in the control group engaged in educational materials (eTables 2-7 in Supplement 3).

Effects on Lipid Profile Control

Effects on lipid profile at 6 months are presented in Table 2. The mean (SD) LDL-C level was 76.3 (37.5) mg/dL in the intervention group and 85.7 (37.1) mg/dL in the control group (Table 2); the adjusted mean difference was -6.62 mg/dL (95% CI, -11.11 to -2.13; *P* = .004). LDL concentration less than 100 mg/dL was observed in 496 patients (79%) in the intervention group and 503 patients (72%) in the control group.

Figure 1. Consolidated Standards of Reporting Trials (CONSORT) Flowchart



POCT indicates point-of-care low-density lipoprotein cholesterol testing.

The LDL-C relative change in was -17.60% vs -11.94% (adjusted difference, -6.32% ; 95% CI, -11.67 to -0.98 ; $P = .02$). LDL-C concentration less than 50 mg/dL was achieved in 148 of 629 patients (23.5%) in the intervention group vs 94 of 700 patients (13.4%) in the control group (population average OR, 1.86; 95% CI, 1.30-2.65; $P < .001$; ICC, 0.008). A 50% or greater LDL-C reduction occurred in 117 of 629 patients (18.6%) in the intervention group vs 94 of 700 patients (13.4%) in the control group (population average OR, 1.76; 95% CI, 1.27-2.43; $P < .001$). Non-HDL-C results were similar (mean difference, -4.19 mg/dL; 95% CI, -8.85 -0.48; $P = .078$; to convert to millimoles per liter, multiply by 0.0259) (eTable 9 in Supplement 3). Baseline, 3-month, and post hoc thresholds analyses appear in eTables 9 and 10 in Supplement 3.

Effects on Prescription of Lipid-Lowering Treatments

Prescription of intensive and combination lipid-lowering therapy was substantially higher in the intervention group at 6 months (Table 2). Combination LLT was prescribed in 441 of 634 patients (69.6%) in the intervention group vs 211 of 713 patients (29.6%) in the control group (population average OR, 3.93; 95% CI, 1.75-8.82; $P < .001$), and intensive LLT was prescribed in 441 of 634 patients (69.6%) in the intervention group vs 212 of 713 patients (29.7%) in the control group (population average OR, 3.91, 95% CI, 1.74-8.78; $P < .001$; ICC, 0.167). Ezetimibe was prescribed to 449 of 634 patients (70.8%) in the intervention group vs 215 of 713 patients (30.2%) in the control group (population average OR, 4.17, 95% CI, 1.88-9.27; $P < .001$; ICC, 0.16). High-intensity statin prescription was

Table 1. Baseline Characteristics of Participating Patients and Clusters

Characteristics	No. (%)	
	Intervention (13 clusters/ 714 patients)	Control (15 clusters/ 751 patients)
Patient baseline characteristics, No.	714	751
Female	266 (37.3)	305 (40.6)
Male	448 (62.7)	446 (59.4)
Age, mean (SD), y	62.76 (9.86)	61.92 (10.39)
Race and ethnicity		
Black	67 (9.4)	105 (14.0)
East Asian	2 (0.3)	6 (0.8)
Indigenous	2 (0.3)	4 (0.5)
Multiracial	361 (50.6)	193 (25.7)
White	282 (39.5)	443 (59.0)
Marital status (married)	406 (56.9)	423 (56.3)
Blood pressure, mean (SD), mm Hg		
Systolic	129.44 (17.88)	133.14 (20.09)
Diastolic	79.45 (9.90)	79.44 (13.28)
Body mass index, median (IQR) ^a	27.43 (24.55-31.04)	27.77 (25.07-30.92)
Current smoker	101 (14.1)	96 (12.8)
Alcohol consumption (<1 d/wk)	598 (83.8)	596 (79.4)
Physical activity (<1 d/wk)	473 (66.2)	480 (63.9)
Coronary artery disease	642 (89.9)	572 (76.2)
Prior myocardial infarction	538 (75.4)	460 (61.3)
Family history of CAD	192 (26.9)	136 (18.1)
Stroke or TIA	63 (8.8)	188 (25.0)
Peripheral or aortic artery disease	88 (12.3)	72 (9.6)
Hypertension	618 (86.6)	622 (82.8)
Diabetes	312 (43.7)	280 (37.3)
Dyslipidemia	585 (81.9)	596 (79.4)
Atrial fibrillation	21 (2.9)	16 (2.1)
Heart failure	58 (8.1)	77 (10.3)
Valvar disease	9 (1.3)	10 (1.3)
Asthma	16 (2.2)	27 (3.6)
COPD	31 (4.3)	19 (2.5)
Chronic kidney disease	34 (4.8)	24 (3.2)
Depression	46 (6.4)	76 (10.1)
Cognitive impairment	1 (0.1)	3 (0.4)
OSAS	11 (1.5)	12 (1.6)
Cluster baseline characteristics		
No.	13	15
Clinical specialty		
Internal medicine	7 (53.8)	8 (53.3)
Cardiology	2 (15.4)	0
Neurology	6 (46.2)	5 (33.3)
Vascular surgery	9 (69.2)	5 (33.3)
Endocrinology	5 (38.5)	8 (53.3)
Family health strategy	2 (15.4)	3 (20.0)
Others	4 (30.8)	2 (13.3)

(continued)

Table 1. Baseline Characteristics of Participating Patients and Clusters (continued)

Characteristics	No. (%)	
	Intervention (13 clusters/ 714 patients)	Control (15 clusters/ 751 patients)
Predominant clinical specialty		
Cardiology	10 (76.9)	14 (93.3)
Endocrinology	2 (15.4)	0
Neurology	0	1 (6.7)
Internal medicine	1 (7.7)	0
Structured protocol for care and management of patients with dyslipidemia or atherosclerosis	7 (53.8)	8 (53.3)
Teaching hospital	10 (76.9)	7 (46.7)
Cluster status		
Public	7 (53.8)	6 (40.0)
Private	6 (46.2)	9 (60.0)
Prior experience with clinical research	12 (92.3)	15 (100.0)
Volume of patients per mo, median (IQR)	80.00 (50.00-200.00)	80.00 (50.00-100.00)
Health care professionals, median (IQR)	6.00 (5.00-11.00)	6.00 (5.00-8.00)
Baseline LDL-C of center-specific medians, median (IQR)	91.00 (80.00-94.50)	93.00 (81.00-100.00)

Abbreviations: CAD, coronary artery disease; COPD, chronic obstructive pulmonary disease; OSAS, obstructive sleep apnea syndrome; TIA, transient ischemic attack.

^a Calculated as weight in kilograms divided by height in meters squared.

higher in the intervention group (549 of 634 patients [86.6%]) vs the control group (477 of 713 patients [66.9%]; population average OR, 2.34; 95% CI, 1.01-5.44; $P = .048$) (eTable 8 in Supplement 3). PCSK9 inhibitors or siRNA-PCSK9 use was negligible in both groups (intervention, 1 of 633 patients [0.2%] vs control, 2 of 713 patients [0.3%]).

Effects on Clinical Events

Within 6 months, 5-point major cardiovascular events (MACE) occurred in 26 of 714 patients (3.6%) in the intervention group and 19 of 751 patients (2.5%) in the control group (HR, 1.44; 95% CI, 0.80-2.60; ICC = 0.000), with no significant between-group difference (Table 3). Likewise, 3-point MACE occurred in 12 of 714 patients (1.7%) in the intervention group vs 11 of 751 patients (1.5%) in the control group (HR, 1.10; 95% CI, 0.42-2.90, ICC = 0.139). Kaplan-Meier curves are provided in eFigure 2 in Supplement 3. Statin-associated muscle symptoms occurred in 27 of 714 patients (3.8%) in the intervention group vs 26 of 751 patients (3.5%) in the control group across statin-associated muscular symptoms categories, with no meaningful differences (eTable 11 in Supplement 3). No differences in serious adverse events were observed (eTable 12 in Supplement 3).

Subgroup Analyses

The subgroup analyses are shown in Figure 2. Across prespecified subgroups, including age (interaction $P = .26$), sex ($P = .68$),

Table 2. Effect of the SAPHIRE-LDL Quality Improvement Intervention on Lipid Profile and LLTs Prescription at 6 Months^a

	No./total No. (%)		Population average OR (95% CI)	P value	ICC ^b
End point	Intervention (13 clusters/714 patients)	Control (15 clusters/751 patients)			
Primary end point					
LDL-C, mean (SD)	76.25 (37.49)	85.67 (37.12)	−6.62 (−11.11 to −2.13) ^c	.004	0.012
No.	629	700	NA	NA	NA
Secondary end points					
Lipid profile					
LDL-C relative change, mean (SD)	−17.60 (39.44)	−11.94 (36.93)	−6.32 (−11.67 to −0.98) ^c	.02	0.021
No.	629	700	NA	NA	NA
LDL-C <50 mg/dL	148/629 (23.5)	94/700 (13.4%)	1.86 (1.30 to 2.65)	<.001	0.008
LDL-C <55 mg/dL ^d	198/629 (31.5)	136/700 (19.4%)	1.82 (1.32 to 2.53)	<.001	0.011
LDL-C <70 mg/dL ^d	343/629 (54.5)	251/700 (35.9%)	2.13 (1.56 to 2.91)	<.001	0.011
LDL-C reduction ≥50%	117/629 (18.6%)	94/700 (13.4%)	1.76 (1.27 to 2.43)	.001	0.005
Non-HDL-C relative change, mean (SD)	−13.98 (31.92)	−9.73 (31.12)	−4.19 (−8.85 to 0.48) ^c	.08	0.025
No.	615	695	NA	NA	NA
LLTs					
Any	618/634 (97.5)	682/713 (95.7%)	1.62 (0.54 to 4.90)	.39	0.023
Combination	441/634 (69.6)	211/713 (29.6%)	3.93 (1.75 to 8.82)	<.001	0.167
Intensive	441/634 (69.6)	212/713 (29.7%)	3.91 (1.74 to 8.78)	<.001	0.167
Any statins	610/634 (96.2)	676/713 (94.8%)	1.26 (0.56 to 2.86)	.58	0.020
Statins intensity			1.48 (0.69 to 3.16)	.32	0.372
High intensity	549/634 (86.6)	477/713 (66.9%)	NA	NA	NA
Moderate intensity	60/634 (9.5)	196/713 (27.5%)			
Low intensity	25/634 (3.9)	40/713 (5.6%)	NA	NA	NA
Ezetimibe	449/634 (70.8)	215/713 (30.2%)	4.17 (1.88 to 9.27)	<.001	0.161
PCSK9i or siRNA PCSK9	1/634 (0.2)	2/713 (0.3%)	NA	NA	NA
Other LLT	60/634 (9.5)	47/713 (6.6%)	1.72 (0.86 to 3.45)	.13	0.012
Adherence to LLT treatments (≥90%)	367/400 (91.8)	479/521 (91.9%)	1.11 (0.52 to 2.37)	.79	0.018

Abbreviations: HDL-C, high-density lipoprotein cholesterol; ICC, intracluster correlation coefficient; LDL-C, low-density lipoprotein cholesterol; LLT, lipid-lowering treatment; NA, not applicable; OR, population average odds ratio; PCSK9i, monoclonal proprotein convertase subtilisin/kexin type 9 inhibitors; SAPHIRE-LDL, Systematic Assessment of a Pragmatic, Practice-Based, Health Care Intervention for Reducing Elevated LDL; siRNA, small-interfering RNA targeting PCSK9.

SI conversion factor: To convert LDL-C to millimoles per liter, multiply by 0.0259.

^a Combination LLT includes ezetimibe combination (ie, ezetimibe plus statins of moderate or high intensity or PCSK9i or siRNA combination [PCSK9i or siRNA PCSK9 plus statin of low, moderate, or high intensity or PCSK9i or siRNA PCSK9 plus statin plus ezetimibe]). Intensive LLT includes any combination LLT or PCSK9i monotherapy, or siRNA PCSK9 monotherapy. High-dose statins

include atorvastatin 40 to 80 mg or rosuvastatin 20 to 40 mg. Moderate-dose statins include atorvastatin 10 to 20 mg, rosuvastatin 5 to 10 mg, simvastatin 20 to 40 mg, lovastatin 10 to 20 mg, pravastatin 40 to 80 mg, fluvastatin 80 mg, and pitavastatin 2 to 4 mg. Low-dose statins include atorvastatin 5 mg, rosuvastatin 2.5 mg, simvastatin 5 to 10 mg, lovastatin 10 to 20 mg, pravastatin 10 to 20 mg, fluvastatin 20 to 40 mg, and pitavastatin 1 mg. Other LLT include fibrates and fish oil.

^b Estimative from generalized estimation equation model considering group (intervention and control) adjusted by stratification variable from phase 1 (sites' mean LDL-C) and patient's baseline LDL-C measure.

^c Mean difference and 95% CI.

^d Post hoc analysis.

cluster status (public vs private; $P = .16$), multiple vascular beds compromised ($P = .61$), history of stroke ($P = .69$), peripheral arterial disease ($P = .31$), coronary artery disease ($P = .17$), or baseline LDL-C (above vs below the median; $P = .83$). A statistically significant interaction was detected for teaching unit status (interaction $P = .03$), with a larger difference observed in teaching units (mean difference, -9.55 mg/dL; 95% CI, -15.13 to -3.97) compared with nonteaching units (mean difference, -1.79 mg/dL; 95% CI, -6.19 to 2.61) (Figure 2). Intolerance and general

units' subgroups were not assessed due to low frequency and mixed specialties presence, respectively.

Sensitivity Analyses

Prespecified sensitivity analyses were consistent with the primary result. Adjustment for age, unit type, teaching status, and specialty showed a mean difference of -7.42 mg/dL (95% CI, -12.25 to -2.58 ; $P = .003$) (eTable 13 in Supplement 3). Restriction to POCT-derived LDL-C produced a difference of -9.50

Table 3. Effects of the SAPHIRE-LDL QI Intervention on Clinical Events

Clinical event	No. (%)		HR or OR (95% CI) ^a	P value	ICC
	Intervention	Control			
Cardiovascular events in 6 mo					
5P-MACE ^b	26 (3.6%)	19 (2.5%)	1.44 (0.8-2.6)	.23	0.000
3P-MACE ^c	12 (1.7%)	11 (1.5%)	1.1 (0.42-2.9)	.84	0.139
Cardiovascular deaths	9 (1.3%)	2 (0.3%)	4.44 (0.87-22.68)	.07	0.170
Nonfatal stroke or TIA	1 (0.1%)	3 (0.4%)	0.35 (0.03-3.57)	.37	0.152
Nonfatal myocardial infarction	2 (0.3%)	6 (0.8%)	0.35 (0.07-1.74)	.21	0.000
Hospitalization for unstable angina	10 (1.4%)	5 (0.7%)	2.64 (0.68-10.26)	.16	0.315
Coronary revascularization	10 (1.4%)	11 (1.5%)	0.96 (0.41-2.26)	.92	0.001
Total deaths	17 (2.4%)	9 (1.2%)	1.88 (0.73-4.88)	.19	0.138

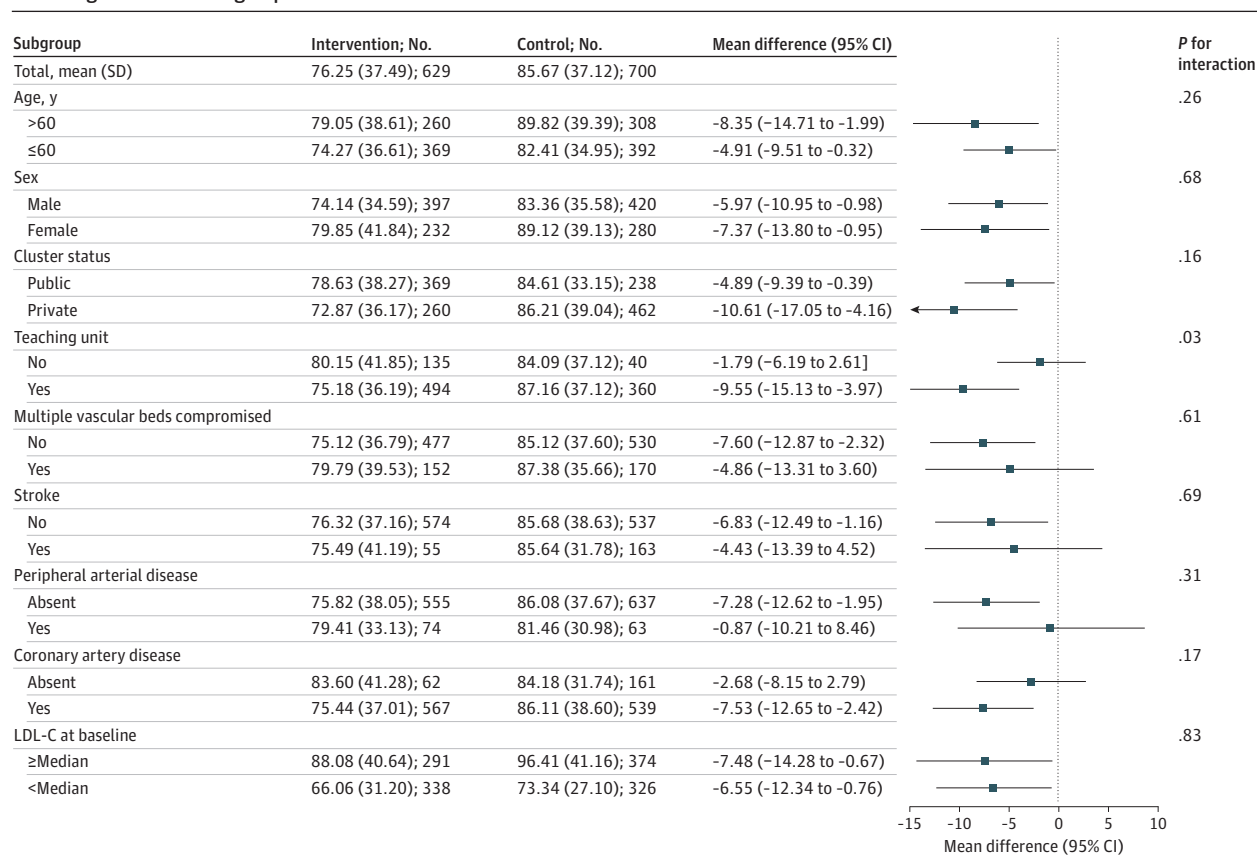
Abbreviations: HR, hazard ratio; ICC, intracluster correlation coefficient; MACE, major cardiovascular events; OR, odds ratio; TIA, transient ischemic attack.

^a Cardiovascular events in 06 months presented as HRs estimates from Frailty Cox proportional hazard models with random effect by cluster.

^b Combined occurrence of a first cardiovascular event (cardiovascular death, nonfatal myocardial infarction, nonfatal stroke or TIA, hospitalization for unstable angina, and coronary revascularization).

^c Combined occurrence of a first cardiovascular event (cardiovascular death, nonfatal myocardial infarction, and nonfatal stroke or TIA).

Figure 2. Forest Plot Showing the Effect of the SAPHIRE-LDL Quality Improvement Intervention on LDL-C Reduction According to Different Subgroups



The prespecified subgroup analysis of participants who tolerated vs could not tolerate statins was not performed due to the low frequency of statins intolerance. The prespecified generalized units vs specialized unit's subgroup analysis was not performed due to the coexistence of multiple specialties in the

unit. LCL-C indicates low-density lipoprotein cholesterol; SAPHIRE-LDL, Systematic Assessment of a Pragmatic, Practice-Based, Health Care Intervention for Reducing Elevated LDL.

mg/dL (95% CI, -14.15 to -4.86; $P < .001$) (eTable 14 in Supplement 3), and calculations using the Martin-Hopkins equation, yielded -7.58 mg/dL (95% CI, -12.50 to -2.66; $P = .003$) (eTable 15 in Supplement 3).

Multiple imputation produced -7.31 mg/dL (95% CI, -12.12 to -2.50; $P = .003$) (eTable 16 in Supplement 3). As all clusters in the intervention group applied the intervention; no per-protocol analysis was performed.

In paired POCT serum measurements available for 109 patients (within 7 days), the Bland-Altman mean (SD) bias was 14.86 (22.35) mg/dL (limits of agreement from -28.94 to 58.67 mg/dL; Pearson $r = 0.81$) (eTable 17 and eFigure 3 in Supplement 3).

Adherence to LLT and Barriers

Adherence to LLT was similar between the intervention group (367 of 400 patients [91.8%]) and the control group (479 of 521 patients [91.9%]). The most common reported barrier for adherence was “difficulty affording the medication.” Additionally, among barriers for prescribing LLT, the most reported was “patient does not have access to high-cost medications” (eTable 18 in Supplement 3).

Discussion

In this pragmatic cluster randomized clinical trial across diverse real-world cardiovascular care settings in Brazil, a digitally enabled, multifaceted QI intervention modestly improved in LDL-C control at 6 months and produced consistent improvements in secondary lipid-related end points, including relative LDL-C reduction, achievement of guideline-recommended LDL-C targets (LDL-C <50 mg/dL), and use of combination therapy.

The observed effect on mean LDL-C was supported by a greater relative reduction and higher likelihood of achieving guideline-recommended targets (<50 mg/dL) and 50% or greater LDL-C reduction. In addition, the observed effect was consistent across multiple prespecified sensitivity analyses, supporting internal validity. Applying the Cholesterol Treatment Trialists' collaboration estimate of approximately 22% relative reduction in major vascular events per 1 mmol/L LDL-C reduction^{13,14} to the observed 0.17 mmol/L between-group difference, the modeled individual effect is modest but potentially population-relevant if sustained and scaled.

A central finding of the SAPHIRE-LDL trial was the shift toward treatment intensification observed in clusters assigned to the QI intervention, notably the prescription uptake of combination LLT including statin-ezetimibe regimens. Overall, statin use was already high at baseline in both groups, indicating that improvements in LDL-C control were driven primarily by uptake of statin-ezetimibe therapy rather than initiation of LLT. This pattern aligns closely with the intervention's design, which integrated real-time clinical decision support (favoring combination therapy) and audit and feedback components—strategies shown in implementation science to influence clinician behavior when coupled with actionable guidance.¹⁵ The early separation in goal-based lipid outcomes at 3 months, followed by a significant difference in mean LDL-C at 6 months, is consistent with a time-dependent process of progressive optimization under routine conditions.

These results should be interpreted in the context of prior attempts to improve lipid management in everyday practice. The Getting to an Improved Understanding of Low-Density Lipoprotein Cholesterol and Dyslipidemia Management (GOULD) registry has demonstrated persistent gaps in LDL-C goal attainment and modest intensification of LLT over time, despite widespread statin use.¹⁶ Similarly, physician or practitioner education-

focused interventions, including the Getting to an Improved Understanding of Low-Density Lipoprotein Cholesterol and Dyslipidemia Management Education (GOULD-EDU) cluster randomized clinical trial, have shown limited impact on treatment optimization and LDL-C control, suggesting that education alone is insufficient to overcome therapeutic inertia.¹⁷ A contemporary cluster randomized clinical trial evaluated a decision support system for early lipid-lowering optimization after acute coronary syndrome, illustrating both the promise and the constraints of single-component interventions.¹⁸ More recently, the Randomized Evaluation of a Remote Management Program to Improve Guideline-Directed Medical Therapy (DRIVE) trial achieved medication prescription in 64% of patients with type 2 diabetes at 6 months using a remote, pharmacist-driven program, although it relied on delegated prescribing authority not available in many health systems, including Brazil.¹⁹ Against this background, SAPHIRE-LDL adds evidence that a multifaceted, digitally enabled approach anchored in an active implementation rationale, integrating decision support with audit and feedback and patient-facing tools, can translate guideline recommendations into measurable improvements in prescribing patterns and lipid outcomes.

The direction of the intervention effect was generally consistent across prespecified patient- and cluster-level subgroups. A statistically significant interaction by teaching status was observed, with a larger LDL-C reduction in teaching centers than in nonteaching units. Although exploratory and requiring cautious interpretation, as observed in previous QI trials, this finding may suggest that structured decision support and feedback tools provide greater incremental benefit in settings embedded in academic resources or established care pathways.

Despite these improvements, only one-quarter of patients in the intervention group achieved guidelines-recommended LDL-C targets within 6 months. These findings indicate that the specific multifaceted implementation strategy and delivery methods evaluated in SAPHIRE-LDL—combining decision support, audit and feedback, clinician education, and patient-facing tools—were only partially effective in closing the residual LDL-C control gap in this high-risk population over a 6-month horizon. This residual gap likely reflects the multifactorial nature of secondary prevention including biological variability, access constraints, and the need for broader adoption of high-potency lipid-lowering strategies and indicates that further refinement of digitally enabled QI methodologies—together with structural improvements in access to advanced therapies—will be needed to fully bridge the evidence-practice gap.

The lack of reimbursement of monoclonal proprotein convertase subtilisin/kexin type 9 influencers in Brazil²⁰ was reflected in the very low use in both groups, illustrating persistent structural barriers to LDL-C target attainment.^{8,11,16,21}

No significant differences were observed in major cardiovascular events were observed at 6 months, consistent with the trial's duration, and prior Brazilian QI trials have demonstrated improvements in evidence-based prescribing without detectable short-term differences in clinical outcomes.^{11,22,23}

To our knowledge, this study was among the first pragmatic cluster randomized clinical trials demonstrating the effect of a nonpharmacological intervention in improving both the prescription of LLT and LDL-C control.

Strengths and Limitations

Several strengths of the trial merit consideration. The pragmatic, cluster randomized design was well-suited to evaluating a complex QI intervention while minimizing contamination between arms. The prerandomization observational phase enabled robust baseline characterization of clusters and informed stratified randomization. The intervention integrated multiple components within a unified, digital platform whose integrated architecture supported consistent deployment across heterogeneous clusters. The inclusion of heterogeneous clinical settings enhances the external validity and prespecified analyses with extensive sensitivity testing support results robustness.

Our trial has some limitations. Most participating sites were specialized cardiovascular clinics, which may limit generalizability to primary care. A prospective cost-effectiveness analysis was not prespecified. The 6-month follow-up was not intended to assess long-term sustainability of LDL-C control or downstream effects on cardiovascular outcomes, given that adherence to LLT might decline over years²⁴; the study was underpowered for clinical events. Although point-of-care LDL-C concentration underestimated paired laboratory values, sensitivity analyses yielded consistent effect estimates and absolute target-attainment proportions in both arms should be interpreted with this measurement consideration in mind;

between-group comparisons remain robust. Although consecutive screening was encouraged and reasons for nonenrollment are reported in Figure 1, the slightly higher proportion of patients declining participation in the intervention arm compared with the control arm suggests potential selection bias toward individuals more receptive to digital tools, which may limit generalizability. The open-label design may have influenced clinician behavior beyond the specific intervention components.

Conclusions

In this cluster randomized clinical trial, a digitally enabled, multifaceted QI intervention embedded within routine clinical care modestly improved LDL-C control and increased use of intensive and combination LLT among patients with established ASCVD. These findings add incremental evidence that a digitally enabled QI strategy can contribute to bridging evidence-practice gaps in secondary cardiovascular prevention and complement pharmacological advances in real-world settings, although the partial effectiveness of the present strategy indicates that further refinement of implementation strategies and digitally enabled delivery methods is needed to fully close the residual care gap.

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Conflict of Interest Disclosures:

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Data Sharing Statement: See [Supplement 5](#).

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